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                   --- Welcome to Maude ---
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           Maude 3.5.1 built: Jul 16 2025 12:00:00
            Copyright 1997-2025 SRI International
                  Sat May 30 16:04:07 2026
```

TESTING SYNTAX

reduce in ST **:** lambda "x" "x" .
rewrites: 0 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Term: lambda "x" "x"

reduce in ST **:** lambda "x" lambda "y" "x" .
rewrites: 0 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Term: lambda "x" lambda "y" "x"

reduce in ST **:** app lambda "x" app "x" "x" lambda "x" app "x" "x" .
rewrites: 0 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Term: app lambda "x" app "x" "x" lambda "x" app "x" "x"

reduce in ST **:** lambda "x" app lambda "y" "y" lambda "z" "z" .
rewrites: 0 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Term: lambda "x" app lambda "y" "y" lambda "z" "z"

reduce in ST **:** app app lambda "x" lambda "y" app app "x" "y" "x" lambda "x" lambda "y" "x" lambda "x" lambda "y" "y" .
rewrites: 0 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Term: app app lambda "x" lambda "y" app app "x" "y" "x" lambda "x" lambda "y" "x" lambda "x" lambda "y" "y"

TESTING SIMPLE TYPE INFERENCE

Omega

reduce in ST **:** *inference*(lambda "x" app "x" "x") .
rewrites: 61 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *error*

Boolean values

reduce in ST **:** *inference*(lambda "x" lambda "y" "x") .
rewrites: 20 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(1) -> t(2) -> t(1))

reduce in ST **:** *inference*(lambda "x" lambda "y" "y") .
rewrites: 18 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(2) -> t(1) -> t(1))

reduce in ST **:** *inference*(lambda "x" app lambda "y" "y" lambda "z" "z") .
rewrites: 103 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(4) -> t(2) -> t(2))

Testing a non-bind variable

reduce in ST **:** *inference*(lambda "x" "z") .
rewrites: 11 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*("z" : t(1), t(2) -> t(1))

SKI COMBINATORS

reduce in ST : *inference*(lambda "x" "x") .
rewrites: 12 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(1) -> t(1))

reduce in ST : *inference*(lambda "x" lambda "y" "x") .
rewrites: 20 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(1) -> t(2) -> t(1))

reduce in ST : *inference*(lambda "x" lambda "y" lambda "z" app app "x" "z" app "y" "z") .
rewrites: 239 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(5) -> t(6) -> t(7)) -> (t(5) -> t(6)) -> t(5) -> t(7))

ANOTHER EXAMPLES

reduce in ST : *inference*(app lambda "x" lambda "y" lambda "z" app app "x" "z" app "y" "z" lambda "a" "a") .
rewrites: 451 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, ((t(6) -> t(7)) -> t(6)) -> (t(6) -> t(7)) -> t(7))

reduce in ST : *inference*(lambda "x" lambda "y" lambda "z" app app "x" "z" "y") .
rewrites: 132 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(2) -> t(4) -> t(5)) -> t(4) -> t(2) -> t(5))

reduce in ST : *inference*(lambda "x" lambda "y" app lambda "z" "x" app "y" "x") .
rewrites: 208 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(7) -> (t(7) -> t(6)) -> t(7))

reduce in ST : *inference*(lambda "x" lambda "y" app app "y" lambda "z" app app "z" "x" "x" "x") .
rewrites: 378 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(8) -> (((t(8) -> t(8) -> t(6)) -> t(6)) -> t(8) -> t(9)) -> t(9))

reduce in ST : *inference*(lambda "x" lambda "y" app app "x" "y" lambda "z" app "y" "z") .
rewrites: 255 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, ((t(5) -> t(6)) -> (t(5) -> t(6)) -> t(7)) -> (t(5) -> t(6)) -> t(7))

FLP NOTES EXAMPLES

reduce in ST : *inference*(lambda "x" lambda "y" app app "x" "y" "y") .
rewrites: 165 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(4) -> t(4) -> t(5)) -> t(4) -> t(5))

reduce in ST : *inference*(lambda "x" lambda "y" lambda "z" app "x" app "y" "z") .
rewrites: 132 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(4) -> t(5)) -> (t(3) -> t(4)) -> t(3) -> t(5))

reduce in ST : *inference*(lambda "x" lambda "y" lambda "z" app app "x" "y" "z") .
rewrites: 132 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(2) -> t(4) -> t(5)) -> t(2) -> t(4) -> t(5))

reduce in ST : *inference*(lambda "x" lambda "y" lambda "z" app app "x" "z" "y") .
rewrites: 132 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(2) -> t(4) -> t(5)) -> t(4) -> t(2) -> t(5))

reduce in ST : *inference*(lambda "x" lambda "y" app lambda "z" "x" app "y" "x") .
rewrites: 208 **in** 4 **ms cpu** (0 **ms real**) (52000 **rewrites/second**)
result PPair: *p*(*nil*, t(7) -> (t(7) -> t(6)) -> t(7))

reduce in ST : *inference*(lambda "x" lambda "y" app app "y" lambda "z" app app "z" "x" "x" "x") .
rewrites: 378 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(8) -> (((t(8) -> t(8) -> t(6)) -> t(6)) -> t(8) -> t(9)) -> t(9))

reduce in ST : *inference*(lambda "x" lambda "y" app app "x" "y" lambda "z" app "y" "z") .
rewrites: 255 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, ((t(5) -> t(6)) -> (t(5) -> t(6)) -> t(7)) -> (t(5) -> t(6)) -> t(7))

Bye.